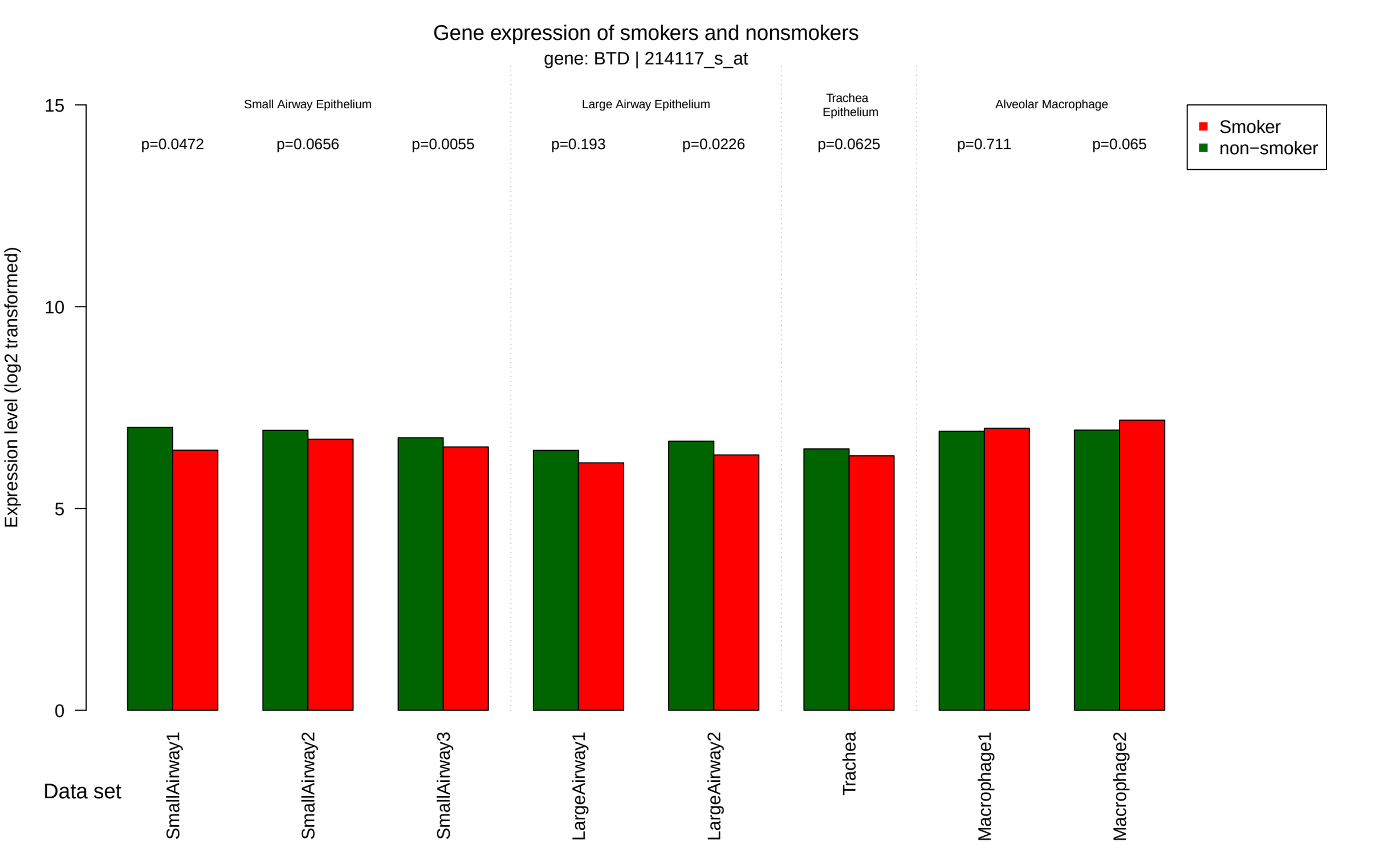
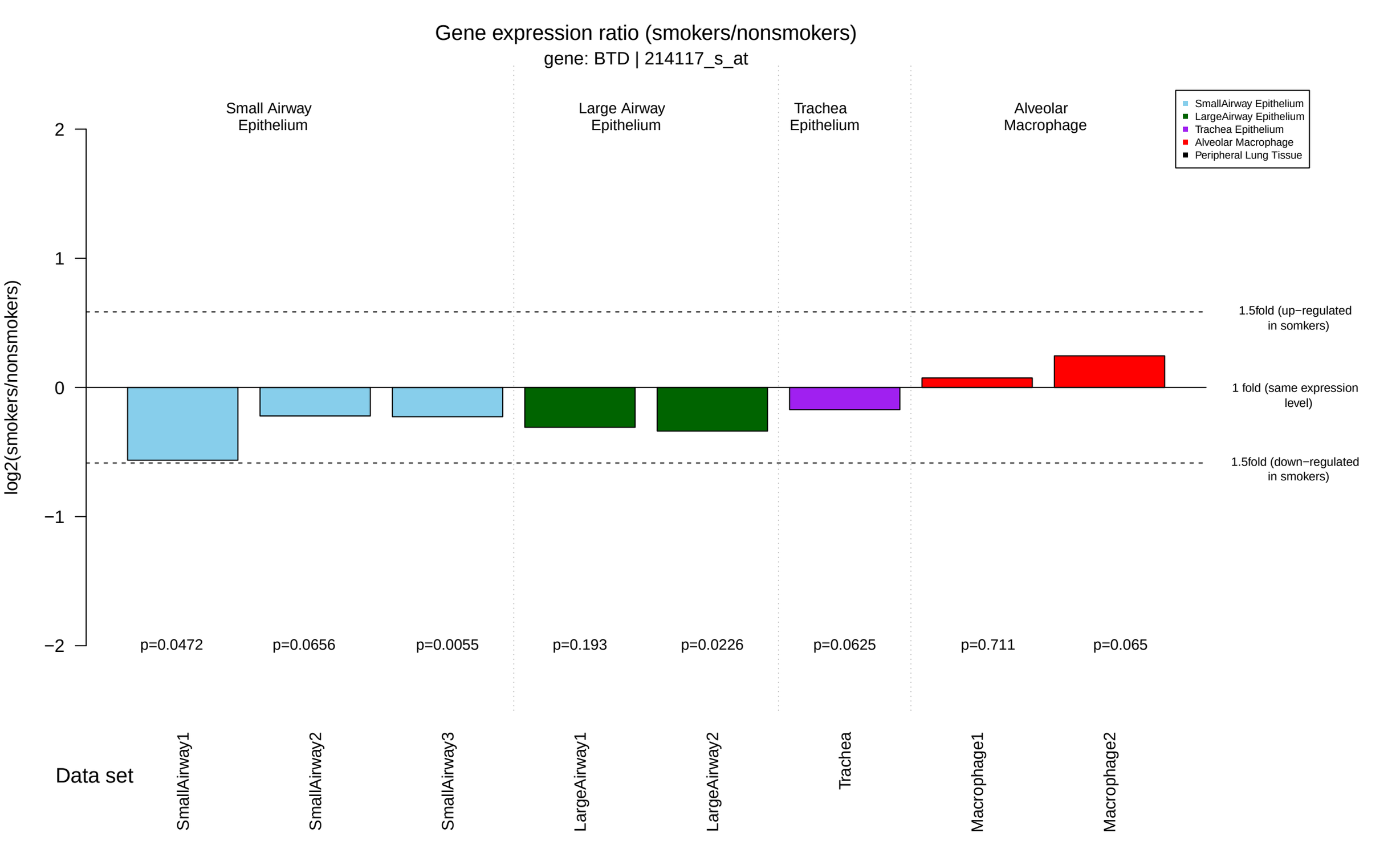
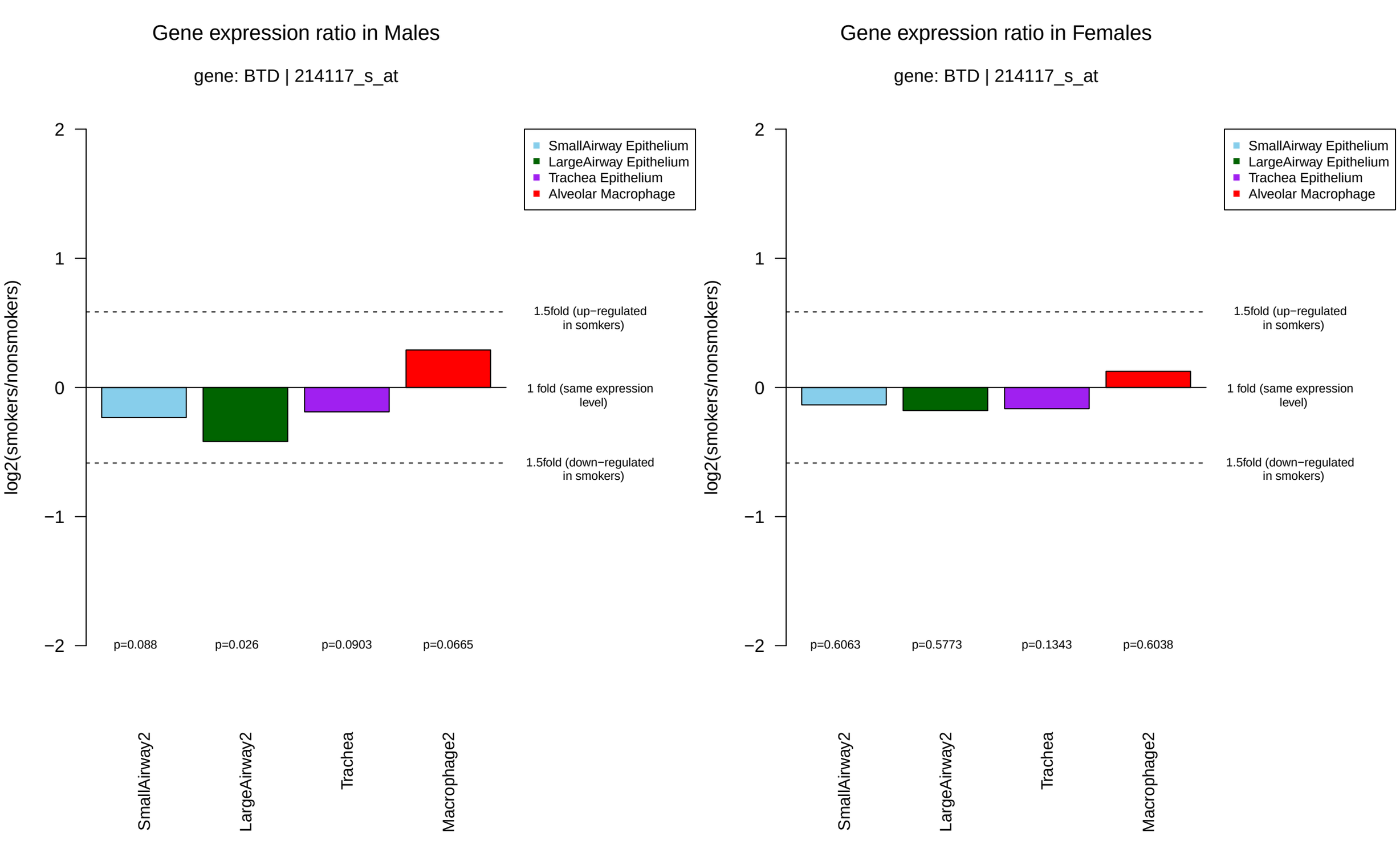
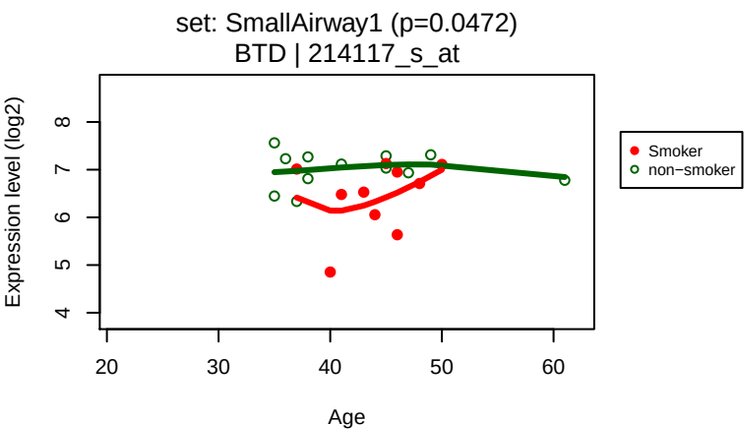
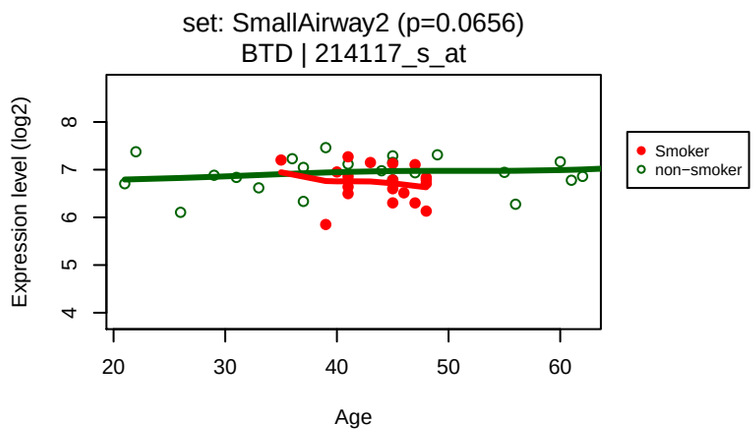
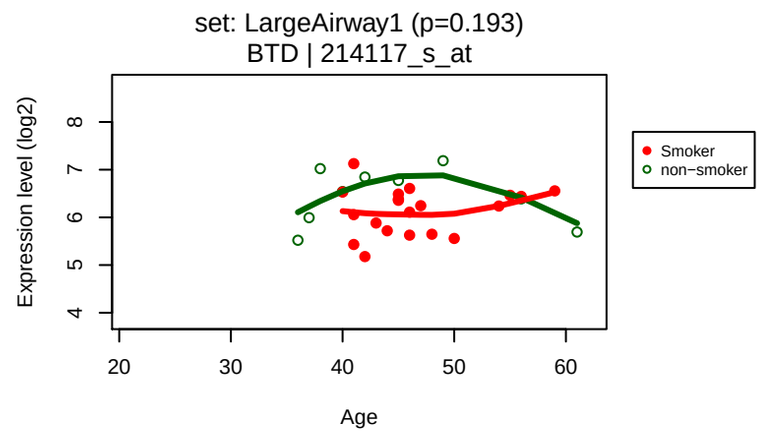

Gene expression of smokers and nonsmokers
gene: BTD | 214117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0472
p=0.0656
p=0.0055
p=0.193
p=0.0226
p=0.0625
p=0.711
p=0.065
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

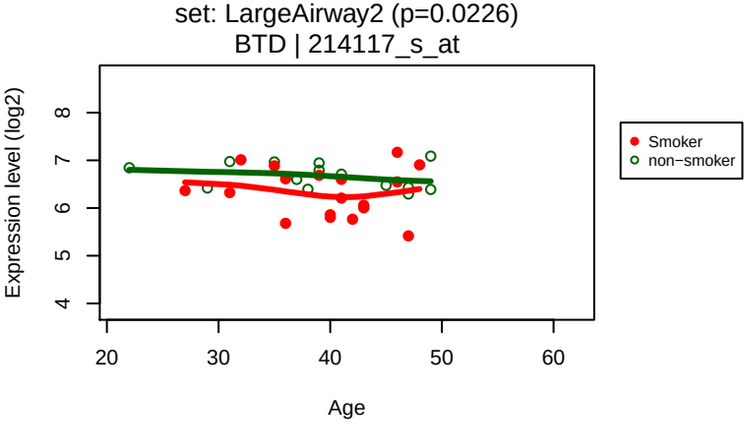

Gene expression ratio (smokers/nonsmokers)
gene: BTD | 214117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0472
p=0.0656
p=0.0055
p=0.193
p=0.0226
p=0.0625
p=0.711
p=0.065
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

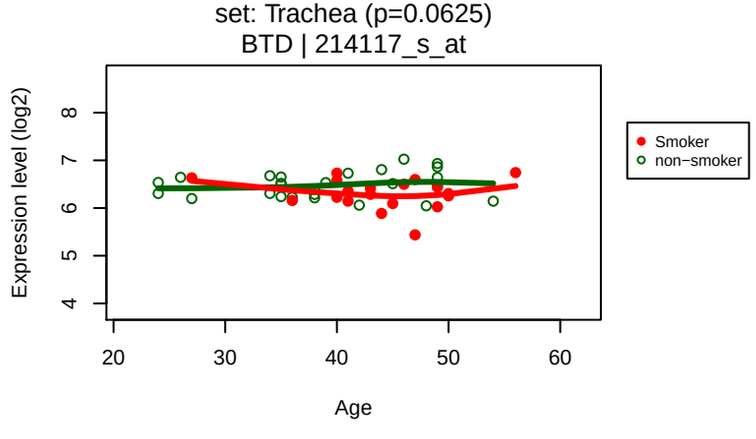

Gene expression ratio in Males
gene: BTD | 214117_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.088
p=0.026
p=0.0903
p=0.0665
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BTD | 214117_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6063
p=0.5773
p=0.1343
p=0.6038
SmallAirway2
LargeAirway2
Trachea
Macrophage2

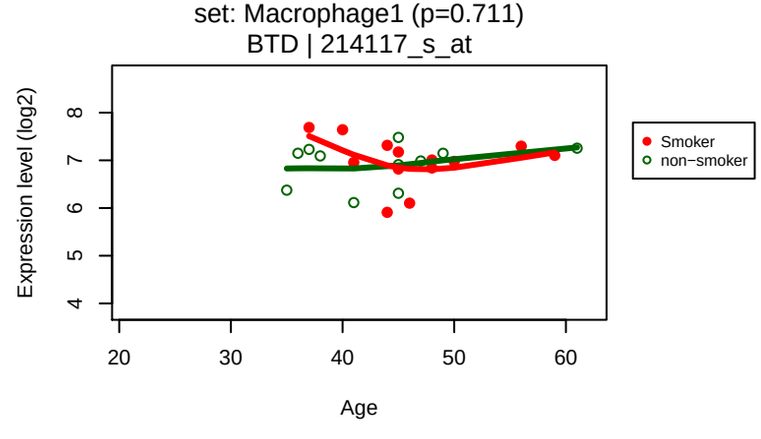

set: SmallAirway1 (p=0.0472)
BTD | 214117_s_at
Expression level (log2)
Age
Smoker
non−smoker

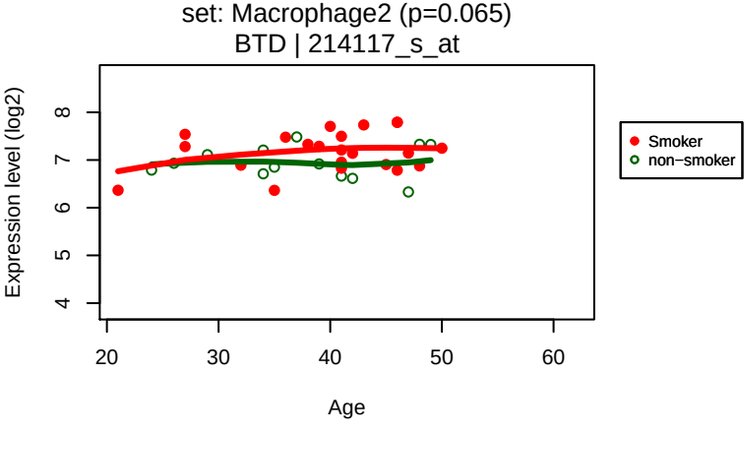

set: SmallAirway2 (p=0.0656)
BTD | 214117_s_at
Expression level (log2)
Age
Smoker
non−smoker

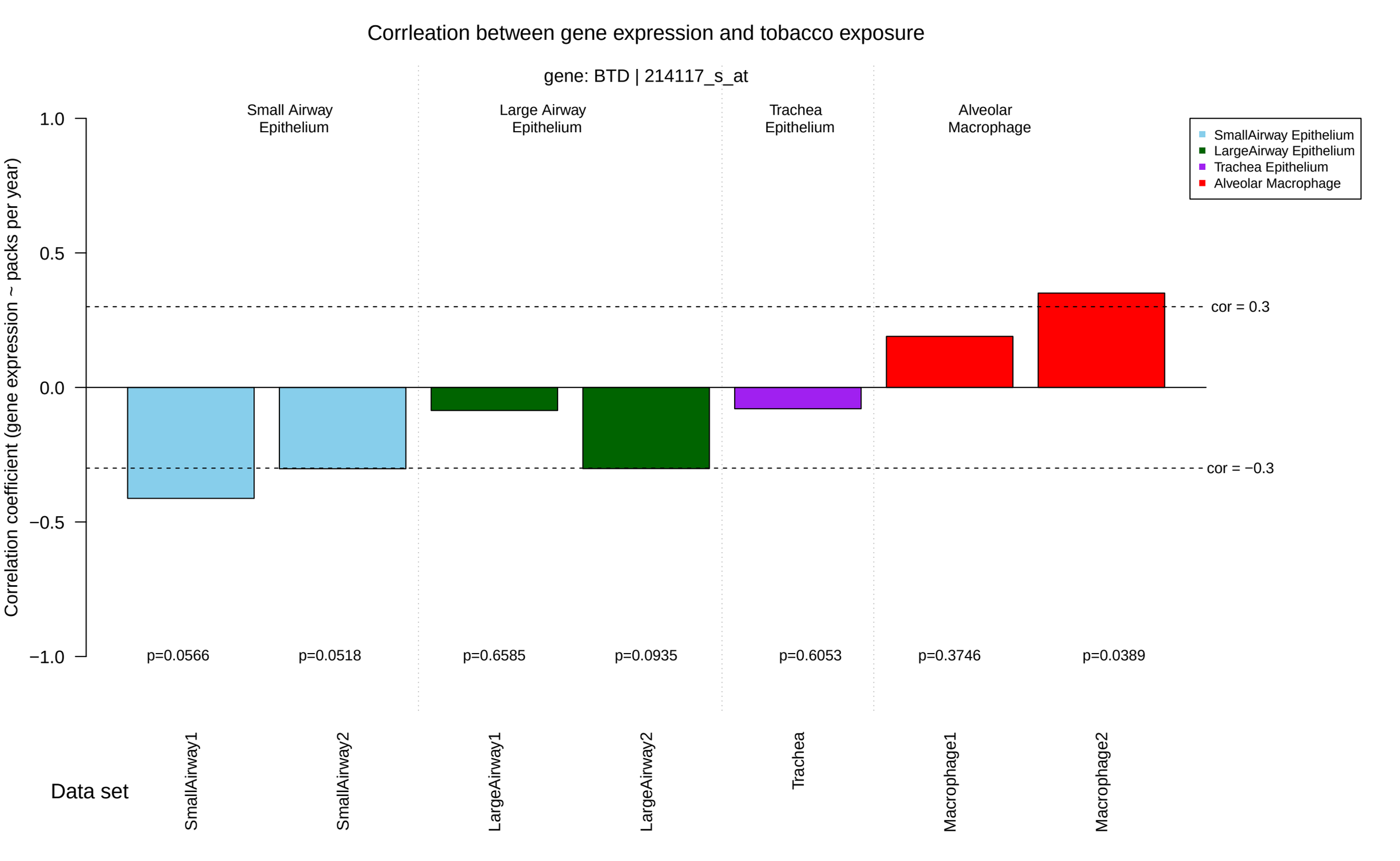

set: LargeAirway1 (p=0.193)
BTD | 214117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0226)
BTD | 214117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0625)
BTD | 214117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.711)
BTD | 214117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.065)
BTD | 214117_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BTD | 214117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0566
p=0.0518
p=0.6585
p=0.0935
p=0.6053
p=0.3746
p=0.0389
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2